

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Designed and built agentic workflows that use LLMs (GPT, Claude, Gemini, Llama) to plan, reason, and execute multi-step tasks, with tool-use and function-calling integrations via the OpenAI Agents SDK and MCP.
- Implemented multi-agent orchestration patterns (LangGraph, CrewAI, AutoGen) for complex task automation, with stateful agent memory via Redis checkpointing at 99%+ uptime.
- Built retrieval-augmented generation (RAG) pipelines on AWS SageMaker to ground agent outputs in proprietary data, serving real-time decisions at sub-200ms p99 latency.
- Established guardrails, evaluation harnesses, and monitoring for agent reliability and task success, blocking 15% of failed releases before production.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Designed, trained, and evaluated computer vision models (CNN and ResNet backbones, UNet-based segmentation, PyTorch) for production image recognition and classification, with model compression and quantization for efficient inference.
- Built and maintained data pipelines for training, validation, and inference, improving RAG retrieval recall 61% to 89% via hybrid dense-sparse retrieval over Pinecone and re-ranking.
- Post-trained and fine-tuned LLMs with LoRA/PEFT, running systematic hyperparameter experiments on multi-GPU clusters.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production microservices in Java, C++, and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS.
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system with ReAct-style reasoning loops automating production incident diagnosis via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed reports.

HoG-ViT

github.com/ganeshhgupta/HoG-ViT

- Built a Vision Transformer incorporating Histogram of Oriented Gradients features into patch embeddings for image classification, improving robustness to lighting and pose variation.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production LLM/agent evaluation framework with rubric-based scoring, drift detection, and regression alerting across model versions; FastAPI backend, PostgreSQL.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Java, TypeScript, C++, SQL

Agentic AI: LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, MCP, Tool-Use/Function-Calling, Multi-Agent Orchestration, ReAct Reasoning

LLM Platforms: OpenAI (GPT), Anthropic (Claude), Google (Gemini, Vertex AI), Prompt Engineering, RAG, LoRA/PEFT Fine-Tuning

ML & Computer Vision: PyTorch, TensorFlow, CNN, ResNet, UNet, Vision Transformers, Model Evaluation, Model Compression/Quantization

Data & Cloud: Vector Databases (Pinecone), PostgreSQL, AWS, GCP, Docker, Kubernetes, CI/CD

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer